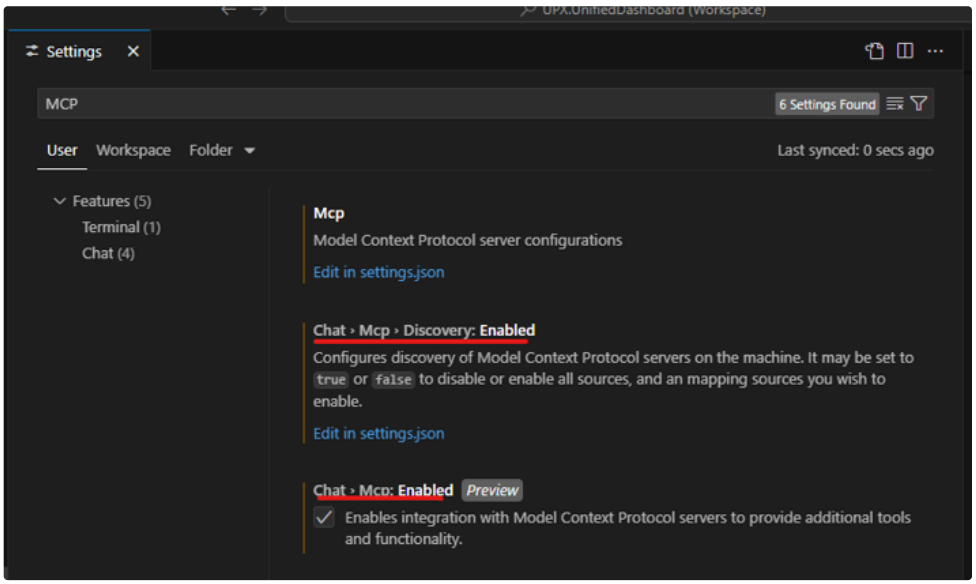
VS Code Setup: MCP (Model context protocol) Configuration

Set up your first MCP with VS Code

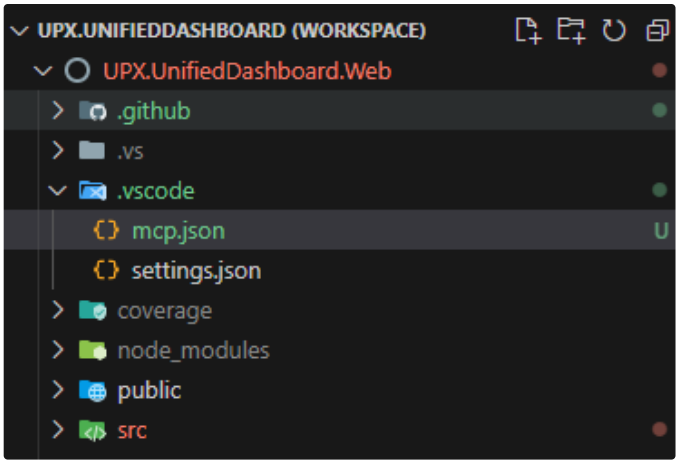
NOTE: The use of external third party MCP server has not yet been approved by the AI COE team. They are currently evaluating this section of Copilot and for other AI coding tools. Please use this solely for learning purposes unless directed otherwise by your leadership.

Prerequisites: Ensure that Github Copilot is functioning properly and free of any errors.

1. To get started, make sure to install Visual Studio Code (VSCode) if you haven't done so yet. You can easily download it from the official website.
2. Ensure you have the following prerequisites ready:
   a. A valid GitHub Copilot license. Link to request for GitHub Copilot
   b. The latest version of Node.js installed.
   c. Finally, ensure you have Visual Studio Code installed.
3. To enhance your experience in VS Code, please make sure the following settings are activated:
   a. Open the VS Code Settings by pressing **Ctrl + ,**.
   b. In the search bar, type **MCP**.
   c. Confirm that both **MCP** and **MCP discovery** are enabled.
   d. 

4. Set Up Your Workspace:
   a. Begin by opening your Git project and confirming that you are in the root directory
   b. At the root, create a folder named `.vscode` to store your project settings.

c. Inside the `.vscode` folder, create a file named `mcp.json`.

d. 

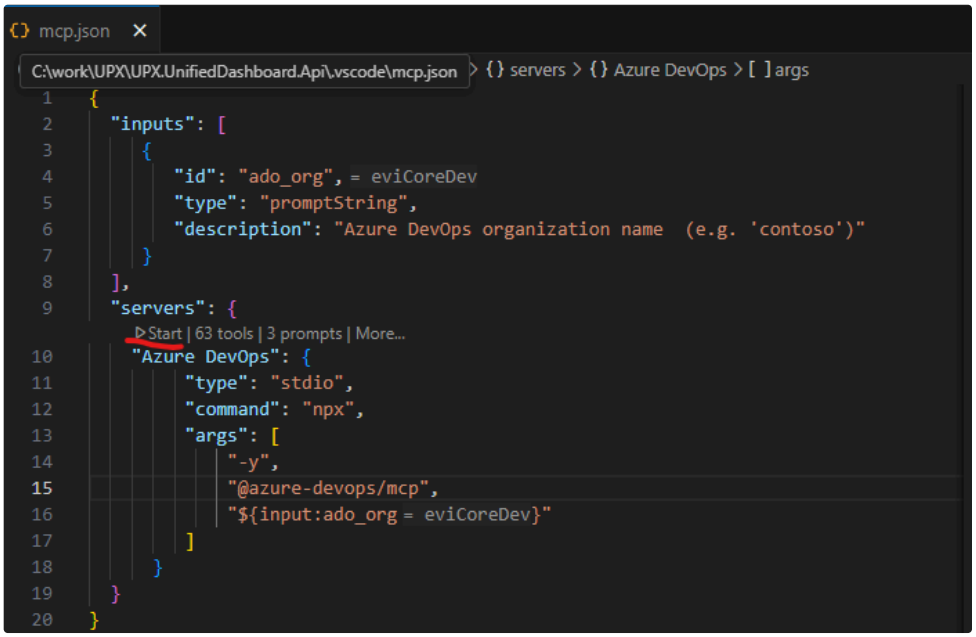
e. Paste the following settings for the Azure DevOps MCP connection:

```
{
   "inputs": [
      {
         "id": "ado_org",
         "type": "promptString",
         "description": "Azure DevOps organization name (e.g. 'contoso')"
      }
   ],
   "servers": {
      "Azure DevOps": {
         "type": "stdio",
         "command": "npx",
         "args": [
            "-y",
            "@azure-devops/mcp",
            "${input:ado_org}"
         ]
      }
   }
}
```

f. Similar to the Azure DevOps MCP, you can configure other available MCP servers. [Refer to this link](#).

g. Please note that these configurations can be applied at either the user or workspace level.

5. To configure your environment, follow these steps:
   - Make sure you have the necessary SDKs or CLI tools installed for the cloud platforms you intend to use (for example, Azure CLI).
   - Utilize the integrated terminal in VSCode to execute any commands required for deploying or managing your MCP setup. You can open the terminal by navigating to `Terminal` > `New Terminal`.
   - If needed, set up authentication for these services. For Azure DevOps, you will need to log in using the `az login` command in the terminal.

6. To verify your setup, please click on "Start," located just above the name of the MCP server configuration.
   - 

   - If Visual Studio fails to start, try restarting it once.
7. Be sure to save your changes and commit your setup to version control, such as Git, if you are utilizing it for your project.
8. Few tasks you can perform with MCP using natural language are listed below. [Complete list of available MCP servers on VSCode](#).
   a. Connect to Azure DevOps (details available [here](#)):
      - Pull story details, create user stories, bugs, or features
      - List projects
      - Perform actions on builds, releases, and repositories
      - Manage test plans
      - Execute searches
   b. Connect to Jira (details available [here](#). In beta version)
      i. Use these examples to understand how to interact with Jira:
         - **Search**: “Find all open bugs in Project Alpha.”
         - **Create/update**: “Create a story titled ‘Redesign onboarding’.”
         - **Bulk create**: “Make five Jira issues from these notes.”
      ii. Known limitations (Beta)

      While the Remote MCP Server provides powerful integration capabilities, please be aware of the following limitations during the beta period:
         - Bulk operations may be constrained by rate limits or may not support all input formats.
         - Custom Jira fields may not be recognized or returned without explicit setup.

    - Some clients like GitHub Copilot and Cursor may require custom configuration or are not fully supported.
    - Workspace or site switching is not currently available within a single session.

c. Connect to Azure and interact with the following services (details available here):
   - 🔎 Azure AI Search
   - ⚙️ Azure App Configuration
   - ☸️ Azure Kubernetes Service (AKS)
   - 📊 Azure Cosmos DB
   - 🧮 Azure Data Explorer
   - 📊 Azure Monitor
   - 🗄️ Azure SQL Database
   - 🔧 Azure Resource Management
   - 💾 Azure Storage

d. Connect with Figma designs (details available here):
   - Generate code from selected frames
   - Extract design context
   - Code smarter with Code Connect

e. Connect with Playwright (details available here):
   - Automate browsers and conduct end-to-end tests
   - Execute user interactions in natural language, such as clicking buttons, entering text, and navigating to URLs
   - Functionality is divided into high-level operations:
     - Core automation
     - Tab management
     - Browser installation
     - Coordinate-based (opt-in via --caps=vision)
     - PDF generation (opt-in via --caps=pdf)

f. Connect with Microsoft documentation (details available here):
   - Search through Microsoft's latest official documentation, which may be more detailed or updated than your training data set
   - Perform semantic searches against Microsoft’s official technical documentation
   - Key capabilities:
     - **High-Quality Content Retrieval:** Search and retrieve relevant content from Microsoft Learn, Azure documentation, Microsoft 365 documentation, and other official

Microsoft sources

- **Semantic Understanding:** Use advanced vector search to find the most contextually relevant documentation for any query
- **Optimized Chunking:** Return up to 10 high-quality content chunks (each max 500 tokens), with article titles, URLs, and self-contained content excerpts
- **Real-time Updates:** Access the latest Microsoft documentation as it's published

- Examples:
  - "Give me the Azure CLI commands to create an Azure Container App with a managed identity. **search Microsoft docs**"
  - "Are you sure this is the right way to implement `IHttpClientFactory` in a .NET 8 minimal API? **search Microsoft docs**"
  - "Is gpt-4.1-mini available in EU regions? **search Microsoft docs**"

g. Connect with DeepWiki (details available here):

- Search through documentation related to public repositories
- Examples:
  - deepwiki fetch how can I use gpt-image-1 with "vercel ai" sdk
  - deepwiki fetch how can I create new blocks in shadcn?
  - deepwiki fetch I want to understand how X works